

Q1:

The output is: hey from message1,

getMessage1 does not have a parameter to take from the container so it will print "hey from message1" then it will return "1" to the container.

Q2:

The output is :

hey from message1

hey from message2

reason:

getMessage1 will always be printed before getMessage2 because getMessage1 does not have any arguments, so it will be printed directly. On the other hand, getMessage2 has an argument and a qualifier, so it will go to Qualifier("1") and take the return value but it will not use it.

Q3:

The first output:

hey from message1

hey from message3

hey from message2

reason:

getMessage1 will be printed first because it does not have an argument and does not depend on any other return method. Then getMessage3 will execute it also has no argument and does not depend on any other return method. Finally, getMessage2 will execute because it depends on the return value of getMessage3.

The second output :

hey from message3

hey from message1

hey from message2

reason:

Reason:

getMessage3 will be printed first because it does not have an argument and does not depend on any return method. Then getMessage1 will execute because it also does not have an argument or dependency. Finally, getMessage2 will execute because it depends on the return value of getMessage3.

the third output :

hey from message3

hey from message2

hey from message1

reason:

At first, getMessage3 can execute before getMessage1 because it does not depend on any return method. Then getMessage2 will execute because it depends on the return value of getMessage3. Finally, getMessage1 can execute because it does not depend on any return value from other methods.

Q4:

The first output:

hey from message1

hey from message3

hey from message2

hey from Main controller

reason:

getMessage1 does not depend on any return method, so it executes first. Then getMessage3 executes since it also does not depend on any return method. Next, getMessage2 gets executed because it depends on the return value from getMessage3 which is already executed. Finally, MainController is executed because it depends on getMessage1 which has already been executed.

The second output:

hey from message3

hey from message2

hey from message1

hey from Main controller

reason:

getMessage3 does not depend on any return method, so it executes first. Then getMessage2 runs because it depends on the return value of getMessage3 (which is already executed). Next, getMessage1 executes because it does not depend on any return method. Finally, MainController executes because it depends on the return from getMessage1 which is already executed.

the third output:

hey from message1

hey from Main controller

hey from message3

hey from message2

reason:

getMessage1 does not depend on any return method, so it starts first. MainController then executes because it depends on the return from getMessage1, which is already executed. Next, getMessage3 executes because it does not depend on any return method. Finally, getMessage2 executes because it depends on the return value from getMessage3 which has already executed

The fourth output:

hey from message3

hey from message1

hey from message2

hey from Main controller

reason:

getMessage3 can run first because it has no dependency on any other bean return value. getMessage1 can run next because it also does not depend on any return value. getMessage2 must run after getMessage3 because getMessage2 depends on the return of getMessage3 (it's injected as a parameter), and since getMessage3 is already executed, getMessage2 can now execute. Finally

MainController runs because it depends on the return from getMessage1 (which was already executed).

the fifth output:

The output:

hey from message3

hey from message1

hey from Main controller

hey from message2

reason:

getMessage3 runs first (no dependencies).

getMessage1 runs next (no dependencies).

MainController can run now because it only depends on the return from getMessage1, and getMessage1 has already executed.

getMessage2 runs last it depends on getMessage3 (which already executed), but it is independent of MainController, so it may be instantiated/executed after the controller.

Q5:

Firs output:

hey from message1

hey from message3

hey from message2

hey from Main controller

reason:

getMessage1 starts first Since getMessage1 (it doesn't depend on other return values), it gets initialized. getMessage3 runs next because it also doesn't depend

on any other return value. getMessage2 executes after getMessage3 because it depends on the return from getMessage3. Finally, MainController is executed last since it depends on getMessage2, which is now fully available.

The second output:

hey from message3

hey from message2

hey from Main controller

hey from message1

reason:

getMessage3 runs first because it has no dependency. Then getMessage2 runs because it depends on getMessage3, which already executed.

After that, MainController can be created since it depends on getMessage2.

Finally, getMessage1 because doesn't depend on any other return value.

The third output:

hey from message3

hey from message1

hey from message2

hey from Main controller

reason:

getMessage3 can run first because it has no dependency on any other return value. getMessage1 can run next because it also does not depend on any return value. getMessage2 must run after getMessage3 because getMessage2 depends on the return of getMessage3, and since getMessage3 is already executed, getMessage2 can now execute.

Finally MainController runs because it depends on the return from getMessage2 (which was already executed).

The fourth output:

hey from message3

hey from message2

hey from message1

hey from Main controller

reason:

getMessage3 starts first since it has no dependency.

Then getMessage2 runs because it depends on getMessage3.

getMessage1 executes next because doesn't depend on any other return value.

Finally, MainController runs because it depends on getMessage3 which already executed.